

10-1.1 Hypothesis Tests on the Difference in Means, Variances Known

Null hypothesis: $H_0: \mu_1 - \mu_2 = \Delta_0$

Test statistic:
$$Z_0 = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad (10-2)$$

Alternative Hypotheses

P-Value

Rejection Criterion For for Fixed-Level Tests

$$H_1: \mu_1 - \mu_2 \neq \Delta_0$$

Probability above $|z_0|$ and
probability below $-|z_0|$,
 $P = 2[1 - \Phi(|z_0|)]$

$$z_0 > z_{\alpha/2} \text{ or } z_0 < -z_{\alpha/2}$$

$$H_1: \mu_1 - \mu_2 > \Delta_0$$

Probability above z_0 ,
 $P = 1 - \Phi(z_0)$

$$z_0 > z_{\alpha}$$

$$H_1: \mu_1 - \mu_2 < \Delta_0$$

Probability below z_0 ,
 $P = \Phi(z_0)$

$$z_0 < -z_{\alpha}$$

Introduction

Test on the μ of
NORMDIST σ^2 known σ^2 unknown

Test on the p

Summary

EXAMPLE 10-1 Paint Drying Time

A product developer is interested in reducing the drying time of a primer paint. Two formulations of the paint are tested; formulation 1 is the standard chemistry, and formulation 2 has a new drying ingredient that should reduce the drying time. From experience, it is known that the standard deviation of drying time is 8 minutes, and this inherent variability should be unaffected by the addition of the new ingredient. Ten specimens are painted with formulation 1, and another 10 specimens are painted with formulation 2; the 20 specimens are painted in random order. The two sample average drying times are $\bar{x}_1 = 121$ minutes and $\bar{x}_2 = 112$ minutes, respectively. What conclusions can the product developer draw about the effectiveness of the new ingredient, using $\alpha = 0.05$?

Introduction

Test on the μ of
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Summary

1. **Parameter of interest:** The quantity of interest is the difference in mean drying times, $\mu_1 - \mu_2$, and $\Delta_0 = 0$.
2. **Non hypothesis:** $H_0: \mu_1 - \mu_2 = 0$, or $H_0: \mu_1 = \mu_2$.
3. **Alternative hypothesis:** $H_1: \mu_1 > \mu_2$. We want to reject H_0 if the new ingredient reduces mean drying time.

4. **Test statistic:** The test statistic is

$$z_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where $\sigma_1^2 = \sigma_2^2 = (8)^2 = 64$ and $n_1 = n_2 = 10$.

5. **Reject H_0 if:** Reject $H_0: \mu_1 = \mu_2$ if the P -value is less than 0.05.

6. **Computations:** Since $\bar{x}_1 = 121$ minutes and $\bar{x}_2 = 112$ minutes, the test statistic is

$$z_0 = \frac{121 - 112}{\sqrt{\frac{(8)^2}{10} + \frac{(8)^2}{10}}} = 2.52$$

7. **Conclusion:** Since $z_0 = 2.52$, the P -value is $P = 1 - \Phi(2.52) = 0.0059$, so we reject H_0 at the $\alpha = 0.05$ level

Practical Interpretation: We conclude that adding the new ingredient to the paint significantly reduces the drying time. This is a strong conclusion.

the β -error for the two-sided alternative,

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$$\beta = \Phi \left(z_{\alpha/2} - \frac{\Delta - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \right) - \Phi \left(-z_{\alpha/2} - \frac{\Delta - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \right)$$

For the two-sided alternative hypothesis with significance level α , the sample size $n_1 = n_2 = n$ required to detect a true difference in means of Δ with power at least $1 - \beta$ is

$$n \simeq \frac{(z_{\alpha/2} + z_{\beta})^2 (\sigma_1^2 + \sigma_2^2)}{(\Delta - \Delta_0)^2} \quad (10-5)$$

For a one-sided alternative hypothesis with significance level α , the sample size $n_1 = n_2 = n$ required to detect a true difference in means of $\Delta (\neq \Delta_0)$ with power at least $1 - \beta$ is

$$n = \frac{(z_\alpha + z_\beta)^2 (\sigma_1^2 + \sigma_2^2)}{(\Delta - \Delta_0)^2} \quad (10-6)$$

EXAMPLE 10-3 Paint Drying Time Sample Size

To illustrate the use of these sample size equations, consider the situation described in Example 10-1, and suppose that if the true difference in drying times is as much as 10 minutes, we want to detect this with probability at least 0.90. Under the null hypothesis, $\Delta_0 = 0$. We have a one-sided alternative hypothesis with $\Delta = 10$, $\alpha = 0.05$ (so $z_\alpha = z_{0.05} = 1.645$), and since the power is 0.9, $\beta = 0.10$ (so $z_\beta = z_{0.10} = 1.28$).

Test on the p

Summary

10-1.3 Confidence Interval on the Difference in Means, Variances Known

If $\bar{x}_1$ and $\bar{x}_2$ are the means of independent random samples of sizes n_1 and n_2 from two independent normal populations with known variances σ_1^2 and σ_2^2 , respectively, a $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$\bar{x}_1 - \bar{x}_2 - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \quad (10-7)$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.

**Sample Size for a
Confidence Interval
on the Difference in
Means, Variances
Known**

$$n = \left(\frac{z_{\alpha/2}}{E} \right)^2 (\sigma_1^2 + \sigma_2^2)$$

**One-Sided
Upper
Confidence
Bound**

$$\mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + z_\alpha \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

and a $100(1 - \alpha)\%$ lower-confidence bound is

**One-Sided
Lower
Confidence
Bound**

$$\bar{x}_1 - \bar{x}_2 - z_\alpha \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2$$

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EXAMPLE 10-4 Aluminum Tensile Strength

Tensile strength tests were performed on two different grades of aluminum spars used in manufacturing the wing of a commercial transport aircraft. From past experience with the spar manufacturing process and the testing procedure, the standard deviations of tensile strengths are assumed to be known. The data obtained are as follows: $n_1 = 10$, $\bar{x}_1 = 87.6$, $\sigma_1 = 1$, $n_2 = 12$, $\bar{x}_2 = 74.5$, and $\sigma_2 = 1.5$. If μ_1 and μ_2 denote the true mean tensile strengths for the two grades of spars, we may find a 90% confidence interval on the difference in mean strength $\mu_1 - \mu_2$ as follows:

10-2.1 Hypotheses Tests on the Difference in Means, Variances Unknown

Tests on the Difference in Means of Two Normal Distributions, Variances Unknown and Equal*

Null hypothesis: $H_0: \mu_1 - \mu_2 = \Delta_0$

Test statistic:
$$T_0 = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad (10-14)$$

Alternative Hypothesis

$$H_1: \mu_1 - \mu_2 \neq \Delta_0$$

$$H_1: \mu_1 - \mu_2 > \Delta_0$$

$$H_1: \mu_1 - \mu_2 < \Delta_0$$

P-Value

Probability above $|t_0|$ and probability below $-|t_0|$

Probability above t_0

Probability below t_0

Rejection Criterion for Fixed-Level Tests

$$t_0 > t_{\alpha/2, n_1 + n_2 - 2} \text{ or}$$

$$t_0 < -t_{\alpha/2, n_1 + n_2 - 2}$$

$$t_0 > t_{\alpha, n_1 + n_2 - 2}$$

$$t_0 < -t_{\alpha, n_1 + n_2 - 2}$$

The **pooled estimator** of σ^2 , denoted by S_p^2 , is defined by

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

EXAMPLE 10-5 Yield from a Catalyst

Two catalysts are being analyzed to determine how they affect the mean yield of a chemical process. Specifically, catalyst 1 is currently in use, but catalyst 2 is acceptable. Since catalyst 2 is cheaper, it should be adopted, providing it does not change the process yield. A test is run in the pilot plant and results in the data shown in Table 10-1. Is there any difference between the mean yields? Use $\alpha = 0.05$, and assume equal variances.

1. **Parameter of interest:** The parameters of interest are μ_1 and μ_2 , the mean process yield using catalysts 1 and 2, respectively, and we want to know if $\mu_1 - \mu_2 = 0$.
2. **Null hypothesis:** $H_0: \mu_1 - \mu_2 = 0$, or $H_0: \mu_1 = \mu_2$
3. **Alternative hypothesis:** $H_1: \mu_1 \neq \mu_2$

4. **Test statistic:** The test statistic is

$$t_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

5. **Reject H_0 if:** Reject H_0 if the P -value is less than 0.05.
6. **Computations:** From Table 10-1 we have $\bar{x}_1 = 92.255$, $s_1 = 2.39$, $n_1 = 8$, $\bar{x}_2 = 92.733$, $s_2 = 2.98$, and $n_2 = 8$. Therefore

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(7)(2.39)^2 + 7(2.98)^2}{8 + 8 - 2} = 7.30$$

$$s_p = \sqrt{7.30} = 2.70$$

**Case 2: Test
Statistic for the
Difference in
Means, Variances
Unknown and
Not Assumed
Equal**

If $H_0: \mu_1 - \mu_2 = \Delta_0$ is true, the statistic

$$T_0^* = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

is distributed approximately as t with degrees of freedom given by

$$v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

If v is not an integer, round down to the nearest integer.

If $\bar{x}_1$, $\bar{x}_2$, s_1^2 , and s_2^2 are the sample means and variances of two random samples of sizes n_1 and n_2 , respectively, from two independent normal populations with unknown but equal variances, then a **100(1 - α)% confidence interval on the difference in means $\mu_1 - \mu_2$** is

$$\begin{aligned} \bar{x}_1 - \bar{x}_2 - t_{\alpha/2, n_1+n_2-2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \\ \leq \mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + t_{\alpha/2, n_1+n_2-2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \end{aligned} \quad (10-19)$$

where $s_p = \sqrt{[(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2]/(n_1 + n_2 - 2)}$ is the pooled estimate of the common population standard deviation, and $t_{\alpha/2, n_1+n_2-2}$ is the upper $\alpha/2$ percentage point of the t distribution with $n_1 + n_2 - 2$ degrees of freedom.

Case 2:
Approximate
Confidence
Interval on the
Difference in
Means,
Variances
Unknown Are
Not Assumed
Equal

If $\bar{x}_1$, $\bar{x}_2$, s_1^2 , and s_2^2 are the means and variances of two random samples of sizes n_1 and n_2 , respectively, from two independent normal populations with unknown and unequal variances, an approximate $100(1 - \alpha)\%$ confidence interval on the difference in means $\mu_1 - \mu_2$ is

$$\bar{x}_1 - \bar{x}_2 - t_{\alpha/2, v} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + t_{\alpha/2, v} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \quad (10-20)$$

where v is given by Equation 10-16 and $t_{\alpha/2, v}$ is the upper $\alpha/2$ percentage point of the t distribution with v degrees of freedom.

10-6.1 Large-Sample Tests on the Difference in Population Proportions

Null hypothesis: $H_0: p_1 = p_2$

Test statistic:
$$Z_0 = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{\hat{P}(1 - \hat{P})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \quad (10-35)$$

Alternative Hypothesis	P-Value	Rejection Criterion for Fixed-Level Tests
$H_1: p_1 \neq p_2$	Probability above $ z_0 $ and probability below $- z_0 $. $P = 2[1 - \Phi(z_0)]$	$z_0 > z_{\alpha/2}$ or $z_0 < -z_{\alpha/2}$
$H_1: p_1 > p_2$	Probability above z_0 . $P = 1 - \Phi(z_0)$	$z_0 > z_\alpha$
$H_1: p_1 < p_2$	Probability below z_0 . $P = \Phi(z_0)$	$z_0 < -z_\alpha$

Introduction

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EXAMPLE 10-15 St. John's Wort

Extracts of St. John's Wort are widely used to treat depression. An article in the April 18, 2001, issue of the *Journal of the American Medical Association* ("Effectiveness of St. John's Wort on Major Depression: A Randomized Controlled Trial") compared the efficacy of a standard extract of St. John's Wort with a placebo in 200 outpatients diagnosed with major depression. Patients were randomly assigned to two groups; one group received the St. John's Wort, and the other received the placebo. After eight weeks, 19 of the placebo-treated patients showed improvement, whereas 27 of those treated with St. John's Wort improved. Is there any reason to believe that St. John's Wort is effective in treating major depression?

1. **Parameter of interest:** The parameters of interest are p_1 and p_2 , the proportion of patients who improve following treatment with St. John's Wort (p_1) or the placebo (p_2).
2. **Null hypothesis:** $H_0: p_1 = p_2$
3. **Alternative hypothesis:** $H_1: p_1 \neq p_2$
4. **Test statistic:** The test statistic is

$$z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where $\hat{p}_1 = 27/100 = 0.27$, $\hat{p}_2 = 19/100 = 0.19$, $n_1 = n_2 = 100$, and

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{19 + 27}{100 + 100} = 0.23$$

5. **Reject H_0 if:** Reject $H_0: p_1 = p_2$ if the P -value is less than 0.05.

6. Computations: The value of the test statistic is

$$z_0 = \frac{0.27 - 0.19}{\sqrt{0.23(0.77)\left(\frac{1}{100} + \frac{1}{100}\right)}} = 1.34$$

7. Conclusions: Since $z_0 = 1.34$, the P -value is $P = 2[1 - \Phi(1.34)] = 0.18$, so, we cannot reject the null hypothesis.

Practical Interpretation: There is insufficient evidence to support the claim that St. John's Wort is effective in treating major depression.

10-6.3 Confidence Interval on the Difference in Population Proportions

Introduction

If $\hat{p}_1$ and $\hat{p}_2$ are the sample proportions of observations in two independent random samples of sizes n_1 and n_2 that belong to a class of interest, an **approximate two-sided $100(1 - \alpha)\%$ confidence interval on the difference in the true proportions $p_1 - p_2$** is

$$\begin{aligned} \hat{p}_1 - \hat{p}_2 - z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \\ \leq p_1 - p_2 \leq \hat{p}_1 - \hat{p}_2 + z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \end{aligned} \quad (10-41)$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.